

CECILIA LOPEZ-GAMUNDI

✉ clopezgamundi@seoe.sc.edu ☎ +1 281 222 3270



EDUCATION

Ph.D., Marine Geoscience University of Miami Rosenstiel School of Marine, Atmospheric, and Earth Science <i>The Sediment Budget of the Great Bahama Bank – Insights from Simulations, Satellites, and Seismic</i>	May 2024 Miami, FL
M.S., Geology The University of Oklahoma School of Geosciences, Mewbourne College of Earth and Energy <i>Integrated Geochemistry and Sedimentology of the Wolfcamp B3 & B2 Intervals, Midland Basin, TX</i>	May 2019 Norman, OK
B.S., Geology The University of Texas at Austin Jackson School of Geosciences, College Scholar	Aug 2014 Austin, TX

ACADEMIC APPOINTMENTS

University of South Carolina Assistant Professor • Coastal and carbonate researcher / PI at the School of the Earth, Ocean, and Environment • Coastal environments of the SE US (2026) – non-major coastal resiliency class (>100 students)	Mar 2026 Columbia, SC
Jet Propulsion Laboratory, NASA / Caltech JPL Affiliate Scientist • Continued research in coastal hydrodynamic modeling and radar data applications	Jan 2026 Pasadena, CA
Jet Propulsion Laboratory, NASA / Caltech JPL Postdoctoral Fellow (M. Simard) • Creating hydrodynamic models of coastal wetlands to simulate carbon export • Using radar satellite products to quantify forest biomass, ocean altimetry, and hydroperiod	April 2025 Dec 2025 Pasadena, CA
University of Miami, RSMAS Research Assistant (S. Purkis) • Quantified carbonate sediment transport during fair- and storm-weather conditions • Calibrated hydrodynamic model (SLIM) using remote sensing observations (MODIS) • Applied machine learning to large earth observation datasets (climate reanalysis)	Jan 2019 Dec 2023 Miami, FL
University of Miami, RSMAS Teaching Assistant (S. Purkis, A. Oehlert, & G. Eberli) • Evolution of the Biosphere (2022) – non major Earth history (>100 students) • Earth System History (2021) – lower division Earth history and geology (50 students) • Depositional and Diagenetic Systems (2020) – upper division sedimentology (15 students)	Aug 2020 Dec 2022 Miami, FL
Phillips and Patricia Frost Science Museum IMPACT Lab Coach • Mentored local high school seniors in designing earth science research projects • Helped students develop petrographic, remote sensing, and field method skills • Aided students in their reconstruction of Pleistocene and Holocene sea level	Jun 2019 Aug 2019 Miami, FL
The University of Oklahoma Research Assistant (R.P. Philp) • Calibrated and maintained HPLC & LECO TOC for daily lab operations • Geochemically analyzed biomarkers (GC & GC-MS) for paleoceanographic reconstructions	Aug 2014 May 2016 Norman, OK
The University of Oklahoma Teaching Assistant (G. Holloway & K. Marfurt) • Introduction to Geology (2016) – non major introductory geoscience lab (>150 students) • Geology & Geophysics for Petroleum Engineers (2015) – senior Petrel lab (40 students)	Aug 2015 May 2016 Norman, OK
The University of Texas at Austin Research Assistant (G. Kocurek, D. Mohrig, & D. Brecker) • Collected wind data for aeolian accumulation analysis in 3D printed Jurassic basin • Used $\delta^{18}\text{O}$ and $\delta^2\text{H}$ values for paleotopographic reconstruction of Western Cordillera	Feb 2012 May 2014 Austin, TX

**POLICY
EXPERIENCE**

Department of Energy | Research Experience in Carbon Sequestration (RECS)

- Gained an in-depth understanding of CCUS technology, industry, and policy
- Applied geologic and geochemical techniques to real world sequestration challenges

Dec 2020
Remote

The American Geosciences Institute | Geoscience Policy Intern

- Attended congressional hearings regarding energy and the environment on Capitol Hill
- Provided the Congressional Hazards Caucus (Senate) with critical geohazards information

Jan 2018
April 2018
Washington DC

**INDUSTRY
EXPERIENCE**

Chevron | R&D Earth Scientist, Reservoir Modeling Chapter

- Forward modeled carbonate formation, hydrodynamic transport, and diagenesis
- Extracted geostatistics from forward models and satellite data and applied to subsurface
- Simulated the effects of ancient sea level and climate on carbonate chemistry and growth

March 2024
March 2025
Houston, TX

Chevron | Earth Science Intern, Carbonate Stratigraphy and Reservoir Quality Prediction

- Enhanced computational stratigraphic forward model for carbonates using Python
- Implemented bioturbation index to assess distribution and sand connectivity
- Parameterized biologic responses to environmental stressors to estimate sediment production

May 2021
Aug 2021
Remote

Shell | Geoscience Intern, Shallow Water Campeche Mexico Exploration

- Created comprehensive pressure database for drilling and exploration purposes
- Identified areas of reservoir connectivity and delineated regional pressure zones
- Contextualized pressure zones with structural regimes and sedimentation rates

June 2020
Aug 2020
Remote

BHP Billiton Petroleum | Geoscience Intern, Trion Production/ Mexico Exploration

- Conducted oil-source rock correlations between Perdido Fold Belt and Campeche Basin
- Utilized kinetics data to create 1D basin models and UEP logs of Mesozoic source rocks
- Identified areas of increased source rock thickness due to underlying paleotopography

May 2018
Aug 2018
Houston, TX

BHP Billiton Petroleum | Geoscience Intern, Gulf of Mexico Production Unit

- Utilized multidisciplinary core data and well logs to investigate low resistivity pay signature
- Created turbidite core database comparing static geologic and dynamic reservoir properties
- Related paleotopographically induced sediment gravity flow deceleration to poorer sorting

May 2017
Aug 2017
Houston, TX

BHP Billiton Petroleum | Geoscience Intern, Eagle Ford Fields Study Team

- Characterized Eagle Ford facies sedimentary for production variability investigation
- Utilized seismic attributes and well logs to identify interbedding heterogeneities
- Identified areas of syndepositional faulting and reservoir compartmentalization

May 2016
Aug 2016
Houston, TX

Pioneer Natural Resources | Geoscience Intern, Shale Technology

- Described and collected XRF and Bambino UCS data from Wolfcamp Formation core
- Compared major element distribution with respect to competent / incompetent bedding
- Utilized trace metal and TOC data to refine sequence stratigraphic interpretation

May 2015
Aug 2015
Dallas, TX

TPS Enterprises | Thin Section Preparation Intern

- Prepared and stained petrographic thin sections
- Translated technical papers into Spanish for PEMEX

Jun 2011
Aug 2011
Houston, TX

APPLICABLE SKILLS AND SELECTED COURSEWORK

Programming: Python, HPC, Unix (advanced); R (intermediate); MATLAB, FORTRAN, Cloud Computing (beginner)
Selected Packages: Python - dask, rasterio, GDAL, xarray, uxarray, rioxarray, rasterio, geopandas, pyTMD, holoviews, plotly, matplotlib, numpy, pandas, scikit-learn, seaborn, arrow, h5py; R - RandomForestSRC, parallel, dplyr, corrplot
Satellites & Models: Landsat 7-8, Sentinel-2, MODIS, VIIRS, SWOT, ICESat-2, ERA5, SWAN, TPXO, Telemac, SLIM
Software: ArcGIS, QGIS, ENVI, GlobalMapper, OpenFlow Dionisos, Trinity, Kinex, Petrel, Kingdom, Spotfire, PowerBI
Analytical: IRMS, GC, GC-MS, MS-MS, MSSV-GC, HPLC, XRF, XRD, SEM-EDS, & Bambino UCS Hammer
Selected Coursework: Geophysical Fluid Dynamics, Stable Isotopes in Biogeochemical Cycles, Geomicrobiology, Environmental Isotope Geochemistry, Global Warming, Hydrogeology, Sediment Gravity Flows, Morphodynamics, Carbonate Petrophysics, Seismic Interpretation of Carbonates, 3D Reservoir Modeling, Petroleum Geochemistry, 3D Seismic Interpretation, High-Temperature Geochemistry, Organic Chemistry I-II, & Inorganic Chemistry
Language: English – native / fluent; Spanish – conversationally fluent, technically proficient

PROFESSIONAL DEVELOPMENT AND WORKSHOPS

Institute for Pure and Applied Mathematics (IPAM), UCLA – Mathematics and Machine Learning for Earth System Simulation Workshop Participant	Feb 2026
Society for Sedimentary Geology (SEPM) – Stratigraphic Forward Modeling Workshop Participant	April 2024
Integrated Ocean Drilling Program – Early-Career Scientific Ocean Drilling Workshop Participant	Aug 2023
Software Carpentry + University of Miami – Python Instructor / R Workshop Helper	2022 – 2023
ExxonMobil – Plates to Pores Geophysical Course Participant	April 2018
American Association of Petroleum Geologists – Imperial Barrel Award Team Member	May 2015
ExxonMobil – Bighorn Basin Field Course Participant	July 2014

AWARDS, SCHOLARSHIPS, AND CERTIFICATIONS

SDI Advanced Open Water, Rescue, Nitrox, and Scientific Diver	2026
NSF Travel Fund (SEPM – ISGC)	2024
Software Carpentry Certified Instructor	2023
UMiami Institute of Data Science and Computing Early Career Award	2023
SEPM Student Travel Fund Recipient	2023
UMiami Student Travel Fund Recipient: International Meeting	2023
Glenn Eugene Laskey Memorial Fellowship Fund Recipient	2015 – 2016
AAPG Imperial Barrel Award: Mid-Continent Section – 1st Place	2015
Apache Corporation Scholars Program Scholarship Recipient	2014
David Scott Holland, Sr. Student Excellence Fund in the Geosciences Recipient, UT Austin	2011 – 2014
College Scholars Honors, UT Austin	2011 – 2014
PADI Open Water Diver	2005

PROFESSIONAL ORGANIZATIONS, OUTREACH, AND LEADERSHIP ROLES

Panel Reviewer: NSF GEO, Swiss National Science Foundation	2026
Associate Editor, <i>Sedimentologica</i>	2025 – cont.
Journal Reviewer: <i>Sedimentary Geology</i> , <i>Basin Research</i> , <i>Geo-Marine Letters</i> , <i>The Depositional Record</i> , and <i>Interpretation</i>	2019 – cont.
Frost Science MUVE Dune Restoration Volunteer	2023 – 2024
UMiami Marine Geosciences Outreach Representative	2020 – 2024
International Association of Sedimentologists (IAS)	2019 – cont.
Society for Sedimentary Geology (SEPM)	2019 – cont.
SEPM Nominations Committee	2024 – cont.
SEPM Carbonate Research Group Committee Organizer	2020 – 2023
Conference Chair for ISGC (Carbonates as a Record of Climate Change)	2024
American Geosciences Institute (AGI)	2016 – cont.
Geologic Society of America (GSA)	2013 – cont.
GSA Membership and Fellowship Committee Student Representative	2023
Society of Exploration Geophysicists (SEG)	2018 – cont.
American Association of Petroleum Geologists (AAPG)	2013 – cont.
Technical Program Committee – Carbonates and Evaporites	2026
Conference Chair for IMAGE – Carbonate Depositional Systems / Evaporites	2021 – 2023, 2026
Abstract Reviewer for IMAGE – AAPG & SEG Joint Meeting	2023, 2026
AAPG University of Oklahoma Student Chapter Secretary	2015 – 2016

FIELD EXPERIENCE

Matagorda Peninsula, TX – B. Carlson & S.D. Khan	Aug 2026
Florida Keys, FL – W. Love, I. Romero, & C. Lopez-Gamundi	Jun 2026
Kentucky & Tennessee – D. Barbeau, A. Leier, & C. Lopez-Gamundi	Apr 2026
Brazos River Delta to Galveston Island, TX – E. Scott	Aug 2025
Permian Basin, TX & NM – Chevron (T. Playton, C. Miller, & B. Ryan)	Feb 2025
Eleuthera, Bahamas – C. Lopez-Gamundi	Jun 2022
Eleuthera, Bahamas – IAS (P. Swart, M. Arienzo, & G. Eberli)	May 2022
New Providence & Exumas, Bahamas – C. Roelfsema, Allen Coral Reef Atlas	Mar 2020
Delaware Basin, NM – J.D. Pigott	Nov 2015
Key West, FL – J.D. Pigott	Oct 2015
Hollywood Quarry, AR – R. Slatt	Sept 2014
Bighorn Basin, WY – ExxonMobil	July 2014
Field Camp, Western US – UT Austin	May – Jun 2014
White Rim Canyon, Moab, UT – G. Kocurek	Aug 2013
Uspallata, Argentina – G. Cisterna, A. Sterren, & O. Lopez-Gamundi	May 2013

PUBLICATIONS

1. **Lopez-Gamundi, C.**, Pelckmans, I., Simard, M., Payandeh, A., Christensen, A., Ward, N., and Heberlein, E. (2026) Quantifying lateral carbon export from coastal wetlands using space-borne sensors and hydrodynamic modeling. In: *Defining the Critical Role and Impact of Sedimentology in a Sustainable Future* (Ed. A. Weislogel and M. Tomlinson), SEPM Miscellaneous Publication 26. <https://doi.org/10.2110/sepmmisc.26.174>
2. Lopez-Gamundi, O. and **Lopez-Gamundi, C.** (2026) Basin-scale sequence stratigraphy of the Vivian Formation (Marañón Basin, Peru). *Journal of South American Earth Sciences*, 174, 105999. <https://doi.org/10.1016/j.jsames.2026.105999>
3. Hupp, B. N., Hashim, M. S., Bryant, R., Kinsley, C. W., Villa, A., Yobo, Lucien, N., and Ajayi, J. O., **Participants of the Establishing Early-Career Scientific Ocean Drilling Learning Communities Workshop** (2025) Perspectives on developing a diverse, knowledgeable and inclusive scientific community in the shifting landscape of US scientific ocean drilling. *Earth Science, Systems and Society*, 5. <https://doi.org/10.1144/esss2024-002>
4. **Lopez-Gamundi, C.**, Barnes B.B., Betzler C., Harris P.M., Oehlert, A.M., Eberli, G.P., and Purkis, S.J. (2025) The carbonate sediment budget of Great Bahama Bank – Earth's largest modern platform. *Geology*, 53, 748-752. <https://doi.org/10.1130/G52850.1>
5. Lopez-Gamundi, O., Cisterna, G.A., Sterren, A. F., and **Lopez-Gamundi, C.** (2024) El Paso Formation: A key unit for the correlation of the Carboniferous glaciation in the Calingasta-Uspallata basin of western Argentina, *Journal of South American Earth Sciences*, 143, 105004. <https://doi.org/10.1016/j.jsames.2024.105004>
6. **Lopez-Gamundi, C.**, Barnes, B.B., Bakker, A.C., Harris, P.M., Eberli, G., and Purkis, S.J. (2023) Spatial, seasonal, and climatic drivers of suspended sediment atop Great Bahama Bank. *Sedimentology*, 71, 769-792. <https://doi.org/10.1111/sed.13151>
7. **Lopez-Gamundi, C.**, Barnes, B.B., Bakker, A.C., Harris, P.M., Eberli, G., and Purkis, S.J. (2023) Deciphering the drivers of suspended sediment – Great Bahama Bank. In: *Advances in the Understanding and Interpretation of Carbonates* (Ed. T.E. Playton and P.M. Harris), SEPM Short Course 57. <https://doi.org/10.2110/sepmiscn.057>
8. **Lopez-Gamundi, C.**, Dobbelaere, T., Hanert, E., Harris, P.M., Eberli, G., and Purkis, S.J. (2022) Simulating sedimentation on the Great Bahama Bank – Sources, sinks and storms. *Sedimentology*, 69, 2693-2714. <https://doi.org/10.1111/sed.13020> – *Selected Editor's Picks 2022*
9. Lopez-Gamundi, O. and **Lopez-Gamundi, C.** (2018) Exhumation of a proximal foredeep and associated wedge-top basin evidenced by porosity versus depth trends: The Upper Cretaceous Vivian sandstones in Northwestern Marañón and Santiago basins (Peru). In: *AAPG Memoir 117: Petroleum Basins and Hydrocarbon Potential of the Andes of Peru and Bolivia* (Ed. G.Z. Valcarce, K.R. McClay, and V.A. Ramos), American Association of Petroleum Geologists, 251-269. <https://doi.org/10.1306/13622123M1173531>

THESIS AND DISSERTATION

1. **Lopez-Gamundi, C.** (2024) The Sediment Budget of the Great Bahama Bank - Insights from Simulations, Satellites, and Seismic (PhD Dissertation). University of Miami, 108. <https://scholarship.miami.edu/esploro/outputs/991032024814502976>
2. **Lopez-Gamundi, C.** (2019) Integrated Geochemistry and Sedimentology of the Wolfcamp B3 and B2 Intervals, Midland Basin, TX. (MS Thesis). The University of Oklahoma, 345. <https://shareok.org/handle/11244/319736>

CONFERENCE
EXTENDED
ABSTRACTS,
POSTERS, AND
PRESENTATIONS

1. **Lopez-Gamundi, C.**, Barnes, B., Betzler, C., Harris, P.M., Oehlert, A.M., Eberli, G.P., and Purkis, S.J. (2026) Over or under? The sediment budget of Great Bahama Bank. International Meeting for Applied Geoscience and Energy (IMAGE) 2026. Houston, Texas. August 17-20, 2026.
2. Simard, M., Christensen, A., Denbina, M., Matte, Pascal; Payandeh, A., and **Lopez-Gamundi, C.** (2025) Using SWOT observations to characterize hydrodynamics and hydroperiod in river deltas and estuaries. SWOT Science Team Meeting. Arcachon, France. October 14-17, 2025.
3. **Lopez-Gamundi, C.**, Harris, P.M., Dobbelaere, T., Hanert, E., Barnes, B. B., Bakker, A. C., Eberli, G. P., and Purkis, S. J. (2024) Elucidating Modern Climate Signals on Great Bahama Bank Using Simulations and Remote Sensing, SEPM International Sedimentary Geosciences Congress. Flagstaff, Arizona. May 5-8, 2024.
4. **Lopez-Gamundi, C.**, Barnes, B. B., Harris, P. M., Eberli, G. P., and Purkis, S. J. (2023) Deciphering the drivers of suspended sediment atop the Great Bahama Bank, 36th International Meeting of Sedimentology. Dubrovnik, Croatia. June 12-16, 2023.
5. **Lopez-Gamundi, C.**, Dobbelaere, T., Hanert, E., Harris, P. M., and Purkis, S. J. (2021) From Source-to-Sink: Combining Remote Sensing and Hydrodynamic Simulation to Develop a Sediment Budget for Great Bahama Bank, AAPG Annual Convention and Exhibition 2021. Denver, Colorado. May 23-26, 2021.
6. **Lopez-Gamundi, C.**, Philp, R.P., and Slatt, R. (2020) The influence of sea level and sediment gravity flows on the sedimentation and geochemistry of the Wolfcamp Formation B3 and B2 Intervals, Midland Basin, TX, AAPG Annual Convention and Exhibition 2020. Online/Virtual. Sept 29 - Oct 1, 2020.
7. **Lopez-Gamundi, C.**, Purkis, S.J., Harris, P.M., Dobbelaere, T., and Hanert, E. (2020) Sediment Production and Transport atop Great Bahama Bank: Insights from Fair and Storm-Weather Conditions on the Andros Platform, AAPG Annual Convention and Exhibition 2020. Online/Virtual. Sept 29 - Oct 1, 2020.
8. Lopez-Gamundi, O. and **Lopez-Gamundi, C.** (2018) Transgressive Reworking and Its Impact on Sandstone Porosity Improvement: The Vivian Formation, Marañón Basin, Northern Perú. AAPG Annual Convention and Exhibition. Salt Lake City, Utah. May 20 - 23, 2018.
9. Lopez-Gamundi, O., Cisterna, G., Sterren, A.F., **Lopez-Gamundi, C.**, and Vergel, M.D.M. (2018) Key sequence stratigraphic surfaces along a basin margin: the Late Paleozoic successions of the Calingasta-Uspallata basin, west Argentina. In: *Revista del Museo de La Plata: VII Simposio Argentino del Paleozoico Superior*. Universidad Nacional de La Plata, Argentina. March 26-28, 2018.

OTHER
INVITED
PRESENTATIONS

1. **Lopez-Gamundi, C.** (2025) Resolving Storm and Climate Signals in the Modern Bahamas Using Satellites and Simulations. Department of Earth Planetary & Space Sciences Colloquium. *University of California Los Angeles*.
2. **Lopez-Gamundi, C.** (2025) Elucidating Modern Climate Signals on Great Bahama Bank Using Satellites and Simulations. School of the Earth, Ocean, and Environment Seminar. *University of South Carolina*.
3. **Lopez-Gamundi, C.** (2024) Suspended Sediment atop Carbonate Platforms – Insights from Satellites and Machine Learning. Exploration Geosciences Network Tech Talk. *Chevron*.
4. **Lopez-Gamundi, C.** (2023) Using New Methods to Solve Old Geologic Problems - What Satellites and Simulations Can Tell Us About Storms and Suspended Sediment. Institute Colloquium (Online). *University of Potsdam*.
5. **Lopez-Gamundi, C.** (2023) Deciphering the Drivers Behind Suspended Sediment – Insights from Long-Term Satellite Monitoring of Great Bahama Bank. Carbonate Café (Online). *Chevron*.
6. True, A., **Lopez-Gamundi, C.**, Elmore, A.G., Chirayath, V., Tárano, A., Jones, N., Johnson, S., Orengo, A., Thornton, M., Best-Otubu, C., Soluri, K., Zuidema, P., McDonald, D., Shay, M., Plotnikova, A., and Isma, L. (2023) Diversity in Marine, Atmospheric, and Earth Science (Online). *University of Miami | North Carolina State University*.
7. **Lopez-Gamundi, C.**, Samankassou, E., Matheson, E., and Burgess, P. (2021) Great Debate: Sea Level Rise Drowns Reefs (Online). *Seds Online*.